PART 1:
How can Intel improve its semiconductor's unit sales revenue, keeping up with new technology trends and accelerating its manufacturing process by implementing its new Internal Foundry System (in the IDM 2.0 business model) and saving around 2billion annually by 2030?

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EXECUTIVE SUMMARY

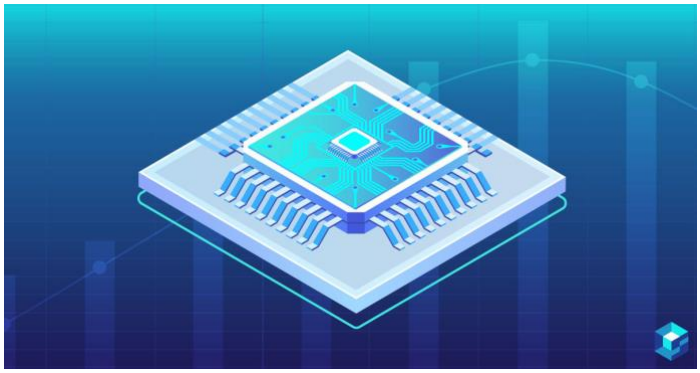
This report advises Intel on how to improve its semiconductor's unit sales, keeping up with new technology trends and accelerating its manufacturing process by implementing its new International Foundry System (in the IDM 2.0 business model) and saving $2 billion annually by 2030. The study is directed specifically to Keyvan Esfarjani, chief global operations officer, the general Manager of the Client Computing Group Michelle Johnston Holthaus, who previously was the chief revenue officer, and to Intel Foundry Services' and Corporate Planning Group's manager Stu Pann. The reader is introduced to the semiconductor industry and receive an explanation of the problem statement. The analysis of Intel current status and inefficiencies in the face of technological improvements will lead to a further exploration of Intel's agility regarding innovation and its sustainability in the future to give the company specific recommendations in the view of a more effective business model.

1. INTRODUCTION

a. Defining the Semiconductor industry

i. What is a semiconductor?

A semiconductor is an electronic device that stores and receives information. It is composed of a die that combine several transistors interconnected that follow different process nodes – specific manufacturing process and design rules. Semiconductors contain different circuit generations relative to the functions they carry. They "are the basic materials for chips" fabrication and "integrated circuits" (IBE, 2023). "A chip is a carrier made of semiconductor material on which multiple electronic components are integrated" (IBE, 2023) as transistors, resistors, and capacitors, used to perform various circuit functions.



(www.sourcengine.com, n.d.)

CPU[1] and GPU[2] are hardware components with different characteristics and usage. A CPU handles the different computing tasks to enable the operating system and applications to run. CPUs are designed for serial instruction processing[3] and have fewer, more powerful cores[4]. They are best suited for general purpose computing applications. A GPU is designed for parallel instruction processing[5], it is more specialized than a CPU. GPUs have more but less powerful cores than CPUs. Currently, they are required for high-performance computing applications such as machine learning and cryptocurrency mining, thanks to their ability to process more data than the CPUs.

Figure 1: Where do we use semiconductors?

(www.sourcengine.com, n.d.)

ii. Moore's law

In 1975, Gordon Moore – co-founder of Intel, predicted the number of transistors on a chip to double every two years. Therefore, increasing computing power while decreasing relative cost. This became the golden rule for electronic innovations that encouraged intel to create faster, smaller, and more affordable transistors for its semiconductors.

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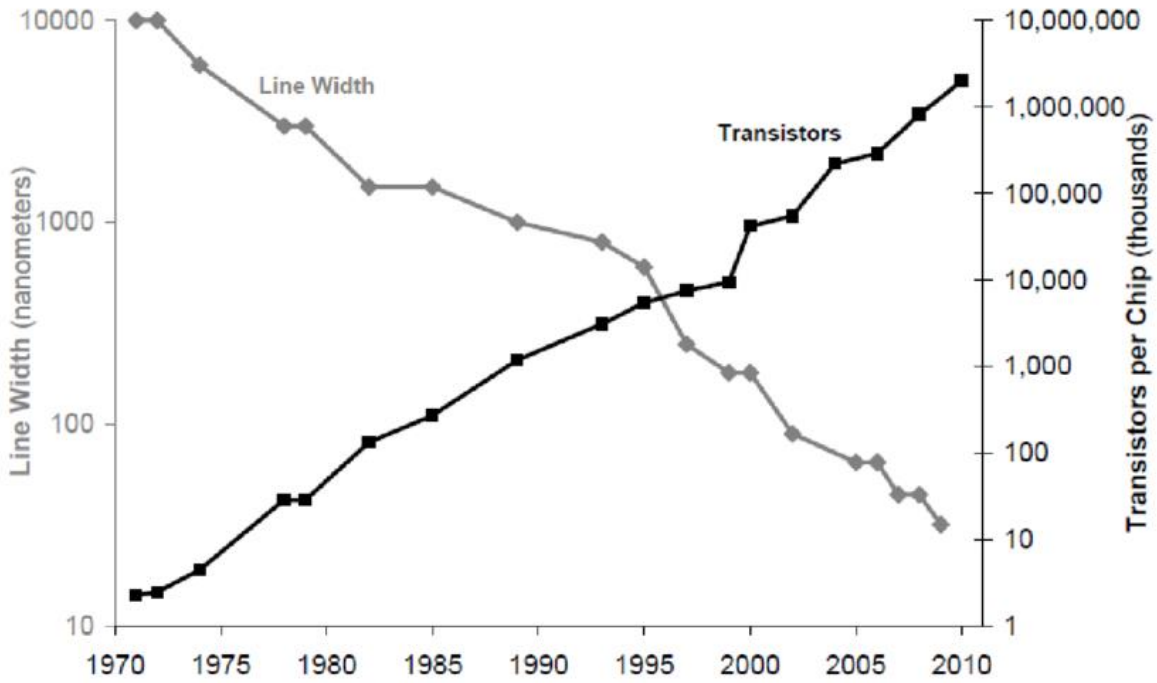
[1] Central Processing Unit
[2] Graphics Processing Unit
[3] Serial processing involves completing one task at a time, in a sequential order. Each task is performed one after the other, and the next task cannot begin until the previous one is completed.
[4] Processing units that handle tasks. In general, a 12-core CPU is faster than a 6-core CPU.
[5] Parallel processing allows a chip to process data simultaneously, hence completing tasks more rapidly.

iii. Challenges of the industry due to Moore's law

Moore's law is not scientific, it relies on observations and is a projection for the future and an objective for semiconductor manufacturers. As infinitely smaller and more powerful semiconductors can only be created to an end many had predicted the end this law. But companies innovated "in other ways, with advanced packaging technology and new materials and the pace of Moore's Law continues". (Intel, 2023)

Figure 2: Intel's chip width and number of transistors per chip over time



(David M. Byrne, 2010)

b. Semiconductor companies' models[6]

Companies in the semiconductor industry are engaged in the design and, or the fabrication of semiconductors and relative devices as transistors and integrated circuits[7]. The three models of semiconductor production are foundries, fabless companies, and IDMs. The last model has the most control over the production process but are the most capital intensive. Fabless companies are more flexible and can react to market changes more quickly but rely on foundries' efficiency at manufacturing their requirements.

i. Foundries

The foundry model consists of semiconductor cutting-edge technology machineries used to create chips designed by fabless firms. TSMC (Taiwan Semiconductor Manufacturing Company) is one of the largest and most advanced foundries. They collaborate with various fabless companies to fabricate cutting edge chips using advanced process nodes. GlobalFoundries is another manufacturing company that offers advanced manufacturing technologies for various industries. Foundries can optimize their fabrication process, by enhancing efficiency and scale.

ii. Fabless company

[6] See appendix 1: The semiconductor industry at a glance.
[7] Also known as microchip that are small electronic device made up of multiple interconnected electronic components as transistors, resistors and capacitors etched onto a tiny piece of silicon.

Fabless firms specialize in chip design, conceptualizing intricate architectures to cater to specific applications (gaming, data processing). Collaboration between foundries and fabless companies is essential to allow physical realization of innovative ideas. NVIDIA is known for graphics processing units (GPUs) used in gaming and AI. The company focuses on designing high-performance chips optimized for specific tasks. AMD (Advanced Micro Devices) is a major competitor of NVIDIA and an important player in the graphics and computing space. Fabless firms focus solely on design without the overhead of manufacturing but are dependent on external foundries resulting in supply chain vulnerabilities and communication challenges between geographically dispersed entities[8].

iii. IDM (Integrated design manufacturer)

IDM companies control all production phases from R&D to chip fabrication and packaging. This vertical integration[9] allows streamlined communication between teams resulting in greater efficiency and faster time-to-market. Intel, Samsung, and Texas instrument are IDM companies with different business model and foundries' strategy. Their tight integration facilitates problem-solving and decision-making, they also have greater control over the manufacturing process which enhances quality assurance. However, they undergo enormous upfront investments for R&D, fabrication facilities, and equipment. Therefore, they have limited specialization in each production phase compared to focused companies.

c. Introduction to Intel

Intel is a global American technology company founded in 1968 that primarily designs and produces computer processors and components for personal computers and servers used in different industries. The majority of their activity relies on the design and manufacture of semiconductors. Intel's products and services are classified under six business units that reflect their various product lines:

The Client Computing Group (CCG) primarily produces PC processors and their components.

Data Centre and Artificial Intelligence Group (DCAI) develops data centre products and drives the company's artificial intelligence (AI) strategy" (Intel Corporation, n.d.).

Network and Edge[10] Group (NEX) acts as a Systems Integrator (SI) for industry verticals offering full suite of silicon, software, and tools for network edge infrastructure[11].

Accelerated Computing Systems and Graphics Group (AXG) designs and delivers GPUs and scalar compute processors and distributed its IP[12] among the CCG and DCAI.

Internal Foundry Services (IFS) is a "fully vertical, standalone foundry business that offers a wide range of manufacturing services to meet unique product needs, including industry-leading sort and test capabilities" (Intel Corporation, n.d.).

Mobileye focus on products for autonomous vehicles.

As the principal revenue of Intel comes from the Client Computing Group, this report will primarily focus the revenue they derive by designing, manufacturing computer chips for their own final products or customers.[13]

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[8] Fabless and foundry companies have different interest regarding their location choice. Fabless need high innovative team usually in developed countries while foundries focus on the material, equipment, and tax expenses.

[9] Vertical integration involves acquisition of a company's supply chain, either upstream or downstream from its own core competency, this model increase control, reduce costs, and improve margins. (NetSuite.com, 2023)

[10] Edge computing's networks and devices process data closer to its source, accelerating and increasing data processing.

[11] A network edge infrastructure is a data structure organized to provision compute edge devices within a network, offsetting processing power requirements from the main servers by allowing the devices to do a majority of the processing work.

[12] The IP (intellectual property) of a semiconductor consists of any specifications, source code, or other information necessary for its manufacture.

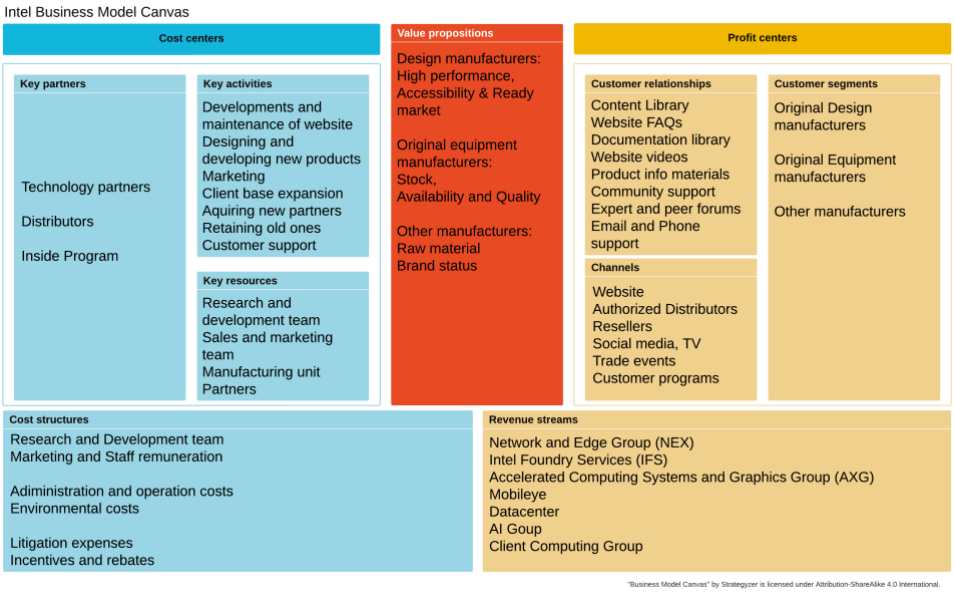
[13] See appendix 5: Intel's Q4 2022 revenue per segment.

Intel “mission statement is: To create world-changing technology that improves the life of every person on the planet”. Intel commits “to make technology inclusive and expand digital readiness for all” (Intel, n.d.). Their strategic priorities are to accelerate growth, improve execution, deploying their stockholder’s capital and evolving their culture.

d. Intel revenue steam

Intel invests heavily in research and development to create cutting-edge technology, giving them a competitive edge in the market. Additionally, they often have patents on their designs, giving them legal protection against direct copying. This means that even if others try to copy their work, they may still face legal obstacles. Additionally, Intel has established close relationships with many computer manufacturers and other partners, giving them significant market share. Ultimately, brand reputation and trust in their products play a key role in customer preferences, often leading them to pay more for Intel-branded chips. These factors together contribute to Intel's ability to maintain a profitable position in the semiconductor industry.

Business model canvas



(Self-made – with Lucid chart)

2. PROBLEM STATEMENT

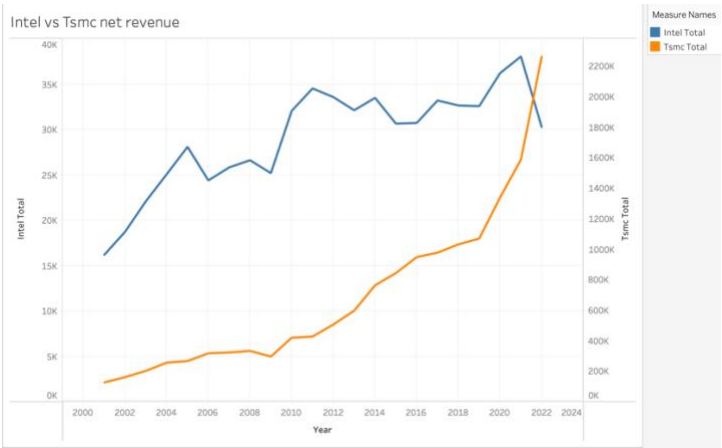
a. Intel current state

Intel enjoyed "over 50 years of dominance in the semiconductor industry"(Greg McFarlane, 2023). However, it now faces strong competitors such as Taiwan Semiconductor Manufacturing (TSMC), Advanced Micro Devices (AMD), and Samsung Electronics, threatening its future in the semiconductor industry. In the last two years Intel has presented a decline in its semiconductor revenue as well as its market share compared to its competitors. Lastly, Intel's revenue decreased by 7.69% in the third quarter of 2023, despite most industry competitors' revenue increase of 9.09%. Its income also fell by 69.58%, slower than its competitors' income growth of 60.02%. Additionally, the overall company revenue decreased by 7.69% and lost market to approximately 7.6%.

b. The semiconductor revenue trend

The graph below shows Intel's semiconductor unit sales relative to time. We notice the company's revenue had a steep increase from 2001 to 2005 and then its progression flattens over the years until 2021 with a pick at $38,084 million and then undergo a serious download trend to $30,281 million in 2022.

Comparing Intel to its rivals TSMC we notice an exact inverse relation between their revenue trends. When Intel has a steep increase at the beginning TSMC is flatter. As time goes, Intel presents slower positive change in revenue to a point where it declines while TSMC has a more rapid and significant revenue escalation to a point where it surpasses Intel's in late 2022.



In the same way, Samsung Electronics surpassed Intel in 2021, Intel taking the second place.

c. Intel management inefficiencies

In 2022, Dell, Lenovo, and HP Inc are the three main customers of Intel. Recently, with the introduction of the machine learning era Intel have been facing important client losses.
In 2020, Apple decided to develop its own semiconductor solutions for its future products. Intel therefore losing a valued and lucrative customer. In its annual report, the company said, “one of its biggest challenges going forward would be in dealing with the loss of revenue from customers like Apple or Microsoft and Amazon that decide to break ties and develop their own semiconductor designs” and send their design to manufacturing in Asia. The principal cause of those shifts raised by the Third Point is Intel’s chip inefficiencies when being implemented in the new technologies as cryptocurrency and AI.

Intel delays in in the production of its 10nm chips[14], disrupted its product roadmap. Also, the company lacked transparency about its inefficiency, not disclosing any explication. Besides, Intel had difficulties integrating EUV[15] technology into its manufacturing process, leading to even more delays in the adoption of smaller nodes. Eventually, when Intel started implementing EUV, he was already significantly behind TSMC, where the new technology was operational to achieve smaller process nodes sooner than Intel.

Moreover, AI[16] new computing trend emerged and presents a challenge for Intel’s CPUs, not well-suited for AI’s specific demands. However, the new parallel processing unit the GPU that runs multiple iterations of the same calculation simultaneously, proved efficiency in AI training systems[17] and gaming. Thus, Intel tried to reach the GPU market, but its chips were useless for the era and failed to satisfy customers. He recently decided to outsource its GPU manufacturing to TSMC. This technological gap affects Intel’s ability to compete effectively in the market as its new chips lack efficiency and performance.

Lastly, Intel’s enclosed culture, unable it to access new external resources, collaborations, and partnerships. Those would have allowed it to follow on new innovations as TSMC did by actively collaborating with researchers to improve its semiconductor technologies. Intel’s reluctance to open up therefore restricted those external collaborations, and partnerships that allow cross-pollination of ideas, ultimately fuelling innovation. The semiconductor market is accelerating, and Intel’s decline comes from an internal problem in the company.

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[14] “nm” refers to the length of a transistor gate, the smaller it is, the more of them can be packed into a single chip, therefore improving the efficiency of the semiconductor. The nanometre is usually a measure used for the identification chip advancement.
[15] Extreme Ultraviolet Lithography enables fabs to achieve smaller process nodes thanks to shorter wavelengths of light that can etch smaller features on silicon wafers.
[16] Artificial Intelligence, implemented into computer systems enable them to perform tasks where human intelligence was needed before.
[17] CPU processes images pixel by pixel, while GPU processes many pixels at once, in this way accelerating the training.

3. MANUFACTURING

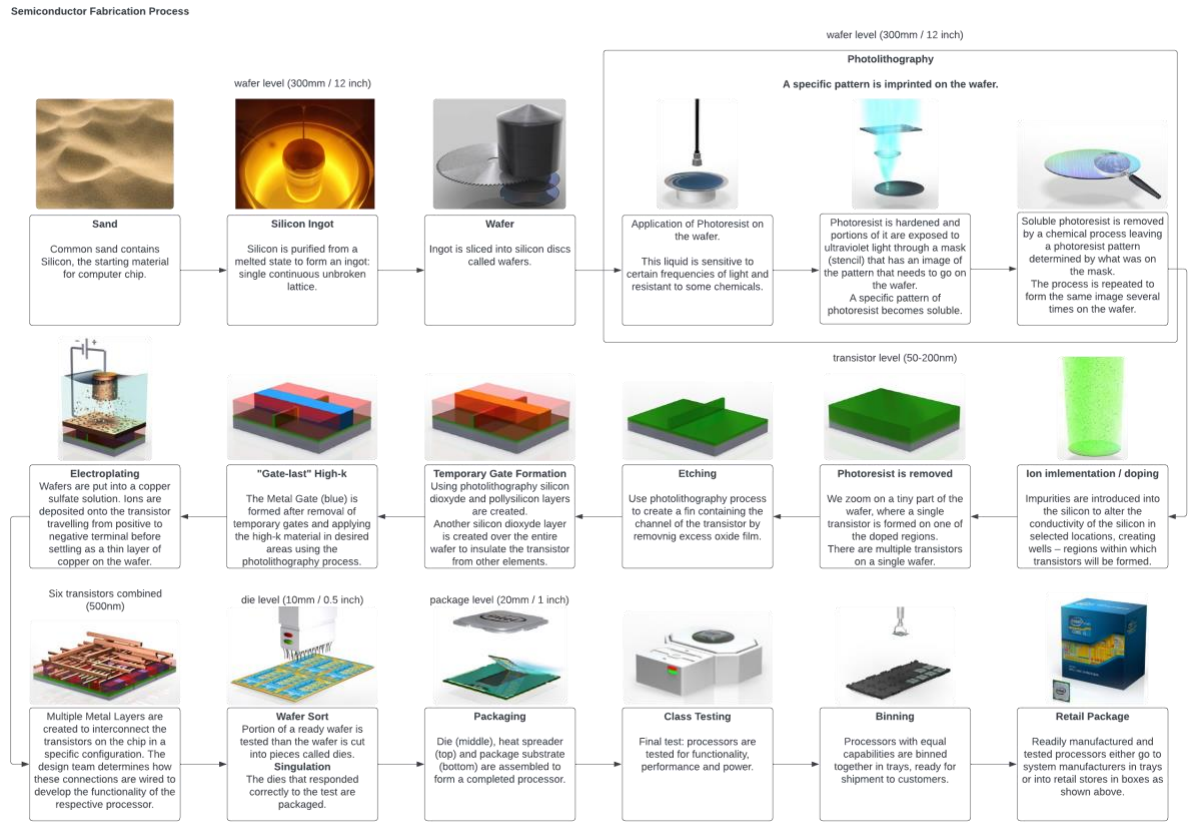
In September 2022, Intel announced a $20 billion project to create new semiconductor manufacturing facility near Columbus in Ohio. The US federal government subsidized the project in billions of dollars to bring manufacturing jobs back to the U.S. and reinforce the country's economic and national security. This is aiming to concur with the People's Republic of China, who becomes the second largest semiconductor exporter in 2022 with a $125 billion exports estimated.

a. Intel semiconductor fabrication process

"A microprocessor has been called the most complex manufactured product made by man." (community.intel.com, 2012)

Intel chips fabrication requires advanced technology and process to follow Moore's Law. Each generation aim to deliver improved functionality, performance, energy efficiency, and lower cost of transistor. The company manufactures chips on a global level through six wafer fabrication sites and four assembly tests.

Figure 3: Intel's semiconductor fabrication (principal steps)



Self-made – with Lucid chart, images from ('Making of a Chip' Illustrations 22nm 3D/Trigate Transistors - Version, 2012) and (BCG Global, 2023))

b. Costs of manufacturing a chip

The term "node generation" refers to the manufacturing technology used to produce the semiconductor chips. A smaller manufacturing process node indicates a more advanced and efficient technology. Each node generation represents a step forward in miniaturization and improvement of semiconductor's performance. The node is identified by the minimum feature size of the chip transistors. For example, a move from a 14nm process to a 10nm process is a transition to a newer node generation. This new product will offer greater precision, smaller transistor sizes, and better performance and energy efficiency. However, developing and implementing more

advanced nodes contribute to the increasing complexity and cost[18] of building and maintaining such facilities. The manufacturing process showed previously of a semiconductor[19] is summarized to the main stages of the process that is usually reiterated many times to create a reliable semiconductor from a new design. Hence, reflecting how complex and costly it can be to produce those devices and the necessary high-tech equipment.

Upfront costs are already high but new factors will accelerate this increase in the following years. The advanced tools needed for the next generation nodes increase equipment outlays. The new class of EUVL machines used to manufacture new chips, cost twice of today's machines that already amounts $150 million. Besides, constructions' expenditure rises as additional equipment need to be housed and production volume maintained to meet more numerous and complex wafer layers[20] requirements. Node advancements also affect process tools used for etching, deposition, and planarization. There is an augmentation of the process steps due to an increase in mask layers' generation. Additionally, strict sustainability regulations imply use of renewable energy. Inflation has also been elevated in the construction sector especially in the US where construction prices raised "by a 6% compound annual growth rate (CAGR) since 2021, and materials by 25% CAGR" (Japan - JP, 2023). Eventually, fabs requires highly specialized complex and gas delivery systems that meet micro-contamination specifications.

To those costs should be added the ongoing expenses of running the facilities that will be driven by the following factors. Greater amounts and more advanced materials are needed to support more complex manufacturing and repair maintenance. Utilities outlays increase due to water, electricity, and natural gas inflation. However as leading-edge fabs are automated and do not need important human intervention, operating labour workforce will remain stable as the principal steps are executed by the machines.

Chips' requirement to be more powerful and small over the years recent semiconductor drive the need for new complex machineries as well as more controlled environment, making fabs extremely expensive to build.

The relationship that gives the fabrication cost per circuit element (transistor) in line with Moore's law is given by:

$$\frac{\$}{\text{element}} = \frac{\dfrac{\$\text{processing cost}}{\text{area "yielded" good silicon chips}}}{\dfrac{\text{elements}}{\text{chip}}} \times \frac{\text{silicon wafer area}}{\text{chip}}.$$

(Corrado et al., 2017)

We can also eliminate the need for assumptions about chip size to obtain a more concise relation:

$$\frac{\$}{\text{element}} = \frac{\$\text{processing cost}}{\text{area "yielded" silicon}} \times \frac{\text{silicon area}}{\text{element}},$$

c. Analyzing Intel manufacturing problem

In semiconductor foundries, wafers are manufactured in lots[21] that goes through the fab's tools preferably maximally utilized to save costs. However, if no care is given to cost and for leading-edge wafers, Intel propose to process them in a few weeks compared to at least two months for the standard process. These are called "hot lots". Intel foundries can do as many of them as the customer wants but they represent important disruption and factory costs impacting its utilization and efficiency for the standard processes. "Our benchmarking suggests that we expedite materials around 2 to 3x more than our industry peers with an estimated 8% to 10% hit in overall output." (Jason Grebe, VP and GM of Intel Corporate Planning Group). The gain from those hot lots must be measured and compared to determine if they are in line with the company's objective.
Designing a chip is expensive, to be produced design is communicated to the fab that after a huge number of process steps, creates the final chip to be tested. Once those tests are received, they are reviewed, and the design is refined multiple times to obtain the finalized semiconductor. Stepping occurs each time a design is sent again to the fab. Intel's customers can ask for "as many stepping as they want" as David Zinsner, Intel CFO said. In

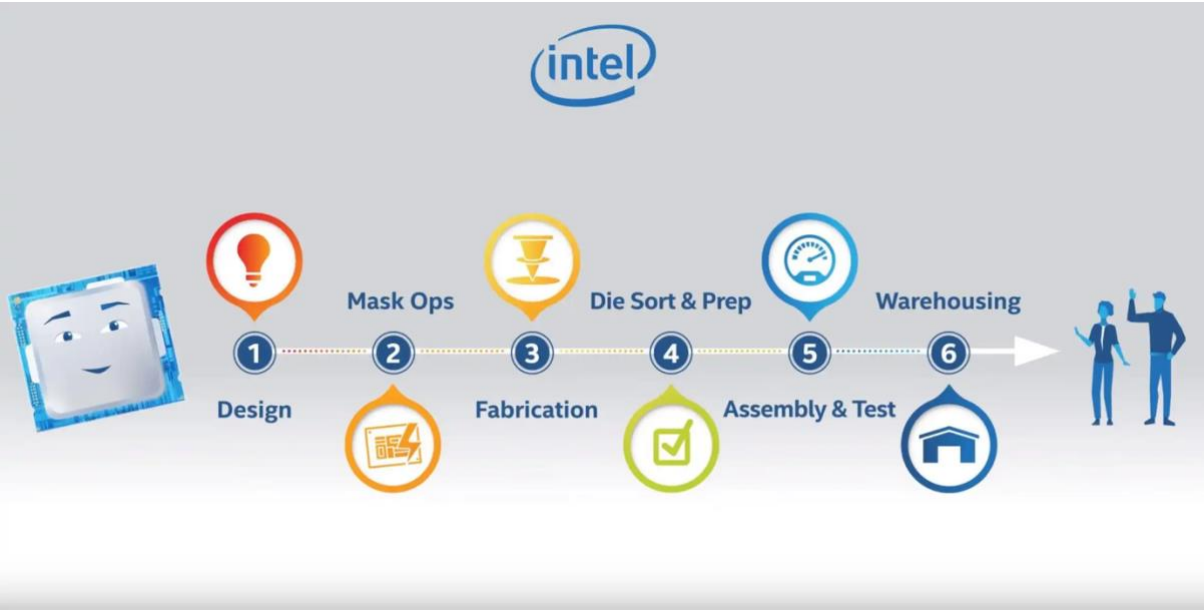
[18] See appendix 7: Advancements in fab technology and increased processing increase future fab costs.
[19] Figure 3
[20] See appendix 6: Layers of a chip illustration.
[21] Also known as batch that are collection of wafers processed sequentially by different tools subjective to their availability.

this way Intel took 12 stepping to market Sapphire Rapids, whereas AMD's took two to three stepping the equivalent Bergamo and Genoa chips. Hence, those excess steps increase Intel's manufacturing costs and delivery time of new design compared to its competitors.

Fabbed chips, are tested and sorted again before being classified into various bins for different products. "We've increasingly grown our test times compared to our industry peers. Currently, we estimate our test times are 2 to 3x those of our competitors. Effectively, our lower-cost testing platform was subsidizing the growth in our test times." claims Jason Grebe, VP and GM of Intel Corporate Planning Group. However, the relevancy of the surplus time spend at this stage is not proven as other companies do not have more issue than Intel with their chips' reliability.

Therefore, Intel's design and manufacturing process present cost and time inefficiencies that require a profound change of its current IDM1.0 business model and should be considered when implementing the new IDM2.0 and Internal Foundry model that we will analyse in part 2.

Figure 4: Intel semiconductor process roadmap



(Intel, n.d.)

4. INNOVATION

Intel margins' fall and delays in time to market prevent it from keeping up with new trends. Therefore, their chips missed the innovations required to be implemented effectively in the use of new technologies as AI. This is the primary reason of their customers' mis contentment and market share losses.

(Hayes, 2022) (Scaled Agile Framework, n.d.) (Team, 2023)

Figure 5: Product life cycle of the semiconductor

(Joyce Jensch, www.questteam.com, n.d.)

The short development cycle and rapid technology change in semiconductor design and manufacture make their life cycle assessment challenging.

Early adopters' attention to new technologies in the semiconductor industry accelerates the introduction stage of innovative products in comparison with the conventional model.[22] Delays could result in important market share losses as customers will choose another provider that meet requirements.

While the "Diffusion of Innovations law"[23] provides insights into the adoption flow, Geoffrey Moore extended this model with his Chasm Theory, introduced in 1991. The chasm represents a gap in the adoption process that occurs between the early adopters and majority. When this phase is passed after the introduction then your product enters the competitive market and can create profit later.

During the growth stage, continuous improvement is pursued on the product as production ramps-up. In, the semiconductor industry production is capacity constrained in materials and equipment. In this phase the semiconductor can be sold at a premium price making companies able to recover investment. Customer are competing on their final product and getting new technologies before competitors represent a leading advantage on the market and helps companies to get positive consumer perception.

"Price on value but charge what the market will bear" as said William H. Davidow, an early Intel VP[24]. This quote introduces the maturity stage, when market share needs to hold for the product to stay at the front, hence pricing is competitive to retain customers and maintain profitability. Semiconductor's equipment typically matures in two to three years following Moore's Law. Photoresist and process chemicals materials usually sustain for four-to-five-years.

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[22] See appendix 2: Standard Product Life Cycle
[23] See Appendix 3: The chasm theory in the product life cycle
[24] Vice President

At the beginning of the decline, offering is very mature, and enhancements can extend the life of the product. Usually, companies rename and reintroduce improved semiconductor versions or research new applications or technologies to differentiate the product. The introduction of better products decreases the value of the old one obliged to be discounted.

The semiconductor company needs now to create a new product that will be competitive with new technologic trends and generate high revenue in its growth phase before again maturing. This is how the cycle works and each time the objective is to create the most revenue generative and sustainable product possible to increase profit and make the company prosper overtime.

That is when Business Agility theory comes to the place. “Agility is the ability to adapt and respond to change, agile organizations view change as an opportunity, not a threat.” (Jim Highsmith, Agile Manifesto). The accelerated life cycle of the semiconductor requires high business agility to allow companies to adapt rapidly to change and especially in the last years with the introduction of new smart technologies as AI, and the Bitcoin that require highly technological chips. (Scaled Agile Framework, n.d.)

5. BUSINESS MODELS

In 2010s, technological evolutions brought several challenges to Intel. The company suffered significant losses in revenue and customer confidence. Those casualties were due to a chaotic design and manufacturing process. To remain competitive on the market, Intel will have to adopt a business model that allows efficient adaptation to technological change.

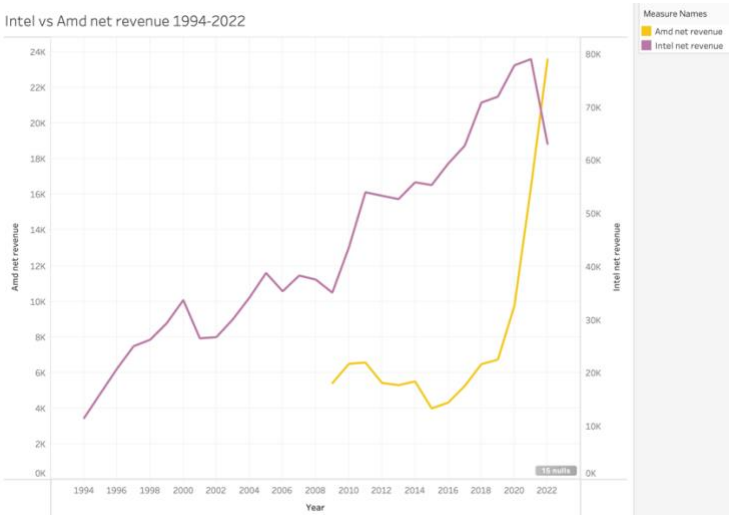
a. Comparing Intel with other companies in the industry

i. Manufacturer: TSMC

TSMC only manufacture chips for their fabless clients. It has been observed over the recent years that foundries can present significant advantages. First and foremost, they undergo less costs than IDMs as they only have to pay for machinery and primary resources. Also, they can price fabless companies for all tests and steps they need to make for the semiconductor design. On second thoughts, a foundry has demand from several different customers and a weakness in the chip design arm will therefore not hit the whole profitability of the company as it would for an IDM who has only one customer. Foundries' diversification enables better utilisation management and higher profits. Moreover, a foundry can also spread technological investments across its several fabless clients. These advantages allowed foundries to take share from IDMs over time and this how TSMC becomes the largest foundry with around 50% market share.

ii. Fabless company: AMD

Fabless companies are not bounded to a single fabrication house, in this way they decide which foundry will better meet their customers' needs. They can focus on the design and innovation of chips without the high cost of operating, maintaining, and upgrading a manufacturing facility. In this way they can more easily drive technological innovation, particularly in areas like mobile communication, consumer electronics, and computing where manufacturing costs are extremely high, as they can swiftly adapt to market changes and demands without the sunk costs of major foundry facilities.
We will use the example of AMD to illustrate the efficiency of the fabless business model. At the beginning in 1970, AMD was fabricating computer chips and became the second-source supplier for Intel before parting away. In 2006, AMD bought ATI for $5.4billion, the major fabless chip company. In 2009, AMD separated its manufacturing parted with its manufacturing unit that becomes the GlobalFoundries and went public in 2021. AMD becoming fabless. "That's when their execution really started to take off because they no longer had to worry about the foundry side of things," (Jay Goldberg, semiconductor consultant at D2D Advisory). AMD now turned to TSMC which makes all of AMD's most advanced chips.
AMD's shift to a fabless model allowed them to focus on innovation, leading to successive architecture updates like Zen+ and Zen 2. This approach, combined with competitive pricing and performance, positioned AMD as a formidable player in the CPU market challenging Intel's dominance. The success of AMD's Ryzen and EPYC processors, coupled with strategic moves in the graphics division, contributed to their resurgence and financial stability. (Evanson and Singer, 2020)



The growth and popularity of fabless companies underscore the numerous benefits they offer. These companies have proven to be nimble and cost-effective, as they eliminate the need for expensive manufacturing infrastructure. They can focus their resources on design and innovation, leading to rapid advancements in technology. The fabless model also allows for greater flexibility, as these companies can swiftly adapt to market changes and trends. Furthermore, by outsourcing production, they can leverage the expertise and scale of dedicated manufacturing firms, ensuring high-quality output and timely delivery. This strategic approach has made them an increasingly dominant force in the tech industry, shaping the future of hardware innovation.

iii. **IDM: Intel**[25]

In the view of Intel manufacturing inefficiency and the success of GlobalFoundries, AMD, TSMC and Samsung and Samsung electronics which all have differing business models compared to Intel. They are completely separated business between fabs and business unit as Samsung, or either only designing as AMD or a manufacturing as TSMC. Those business models make them efficient, flexible, and profitable companies that can adapt quickly to new trends. This last ability is defined as agility which is the most important skill needed in the semiconductor sector today. Intel's lack of agility due to its rigid business model is therefore at root of its decline and market share loss.

"Some companies dominate an industry, fail to innovate, and fall into irrelevance. Others have great ideas but never manage to capitalize on them. The company that can leverage intellectual firepower with commanding market share is the company that can stay both powerful and relevant for decades." (McFarlane, 2019) This is the objective of Intel's new business model IDM2.0 that we will analyse in part 2.

[25] See appendix 8: Intel IDM1.0 versus IDM2.0 (Business Model)

6. OUTLOOK

Part one concentrated, on the problem that leads Intel to a large loss in market share and being destitute of its position as a reference and a leader in the production and design of semiconductors. Intel maintaining its IDM business model seems to have harmed its profitability and efficiency in the production and innovation of semiconductors. In Q3 2021, TSMC took the lead in semiconductor unit sales as a manufacturer and AMD using the same comparison unit as a fabless company also surpasses Intel at the same point in time.
It was noticed fabless and foundry companies respectively AMD and TSMC have been more successful as well as Samsung which have separated the foundry from the design and research and development units[26].

The second part of the report will thoroughly analyse and measure the importance of innovation in the semiconductor industry. The argumentation will consider Intel's unresponsiveness to change as the reason of its failure when technologic change affected the sector. The study will focus on the impact and risk of open or close innovation on business's agility to compare and determine specific recommendations for Intel regarding its business reorganisation to be competitive again in terms of revenue, market share and profitability.

[26] creating distinct companies (Semiconductor Foundry + Memory vendor + OSAT: outsourced semiconductor assembly and test).

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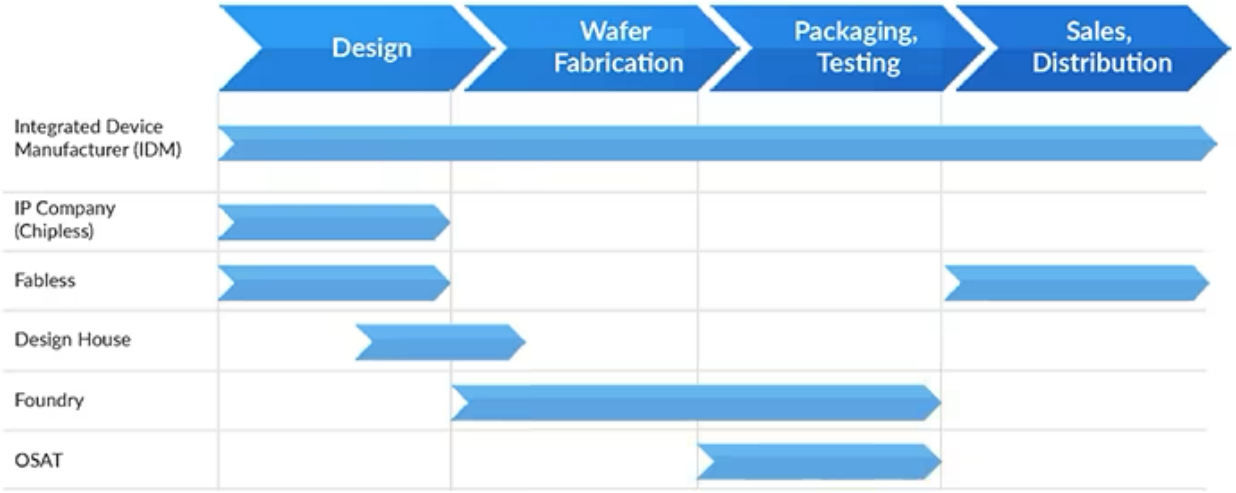
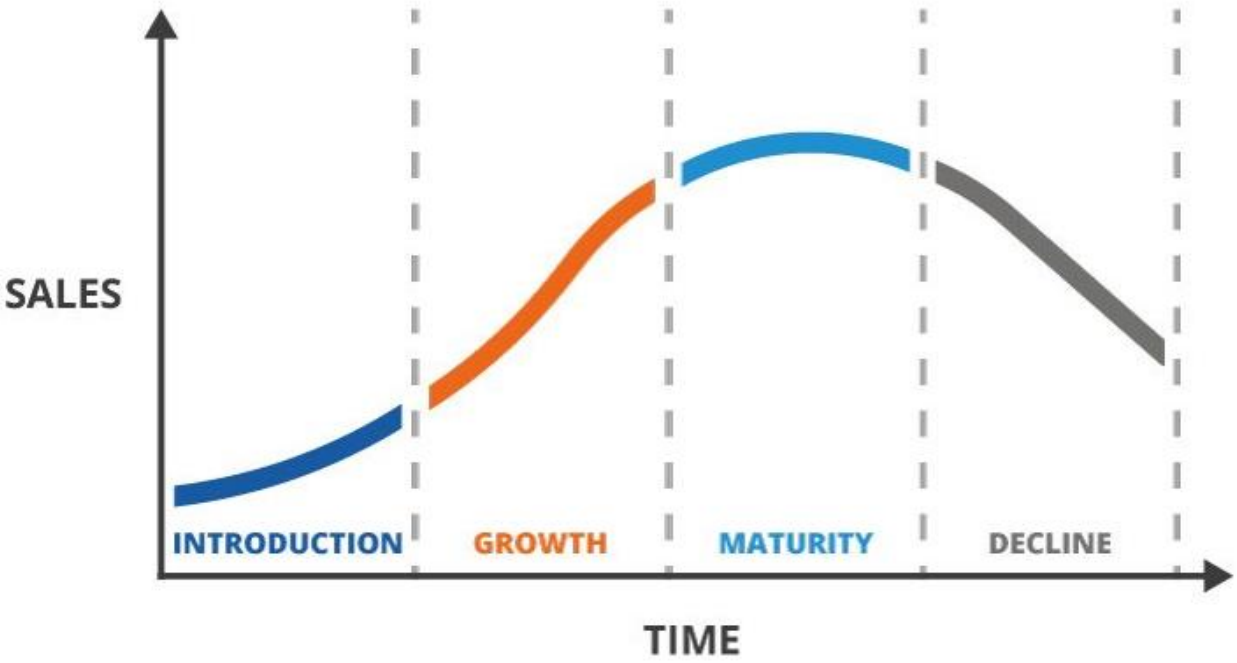
8. APPENDICES

Appendix 1: The semiconductor industry at a glance



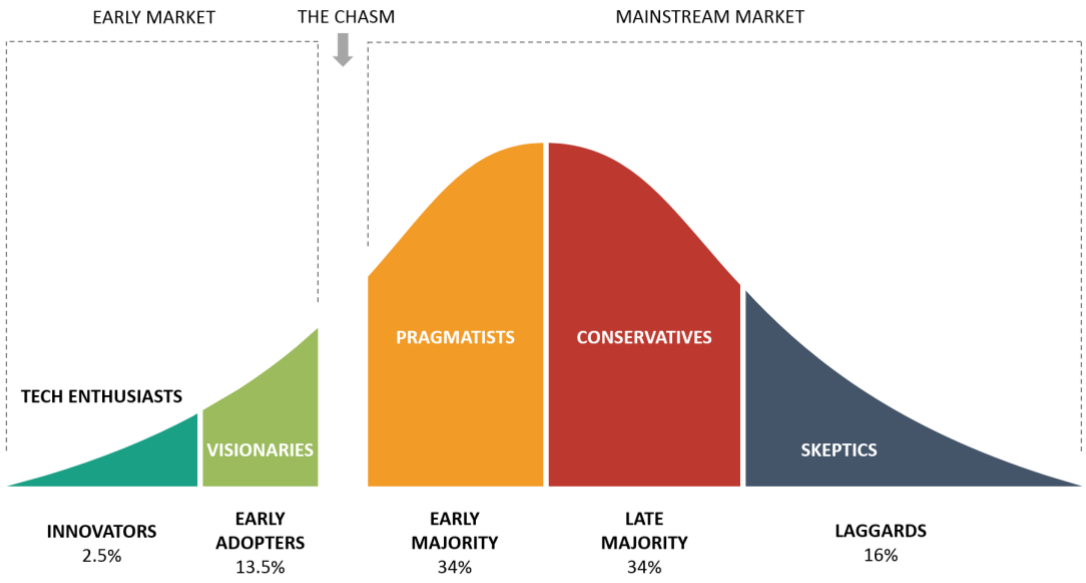
(Samsung Semiconductor EMEA, 2023)

Appendix 2: Standard Product Life Cycle



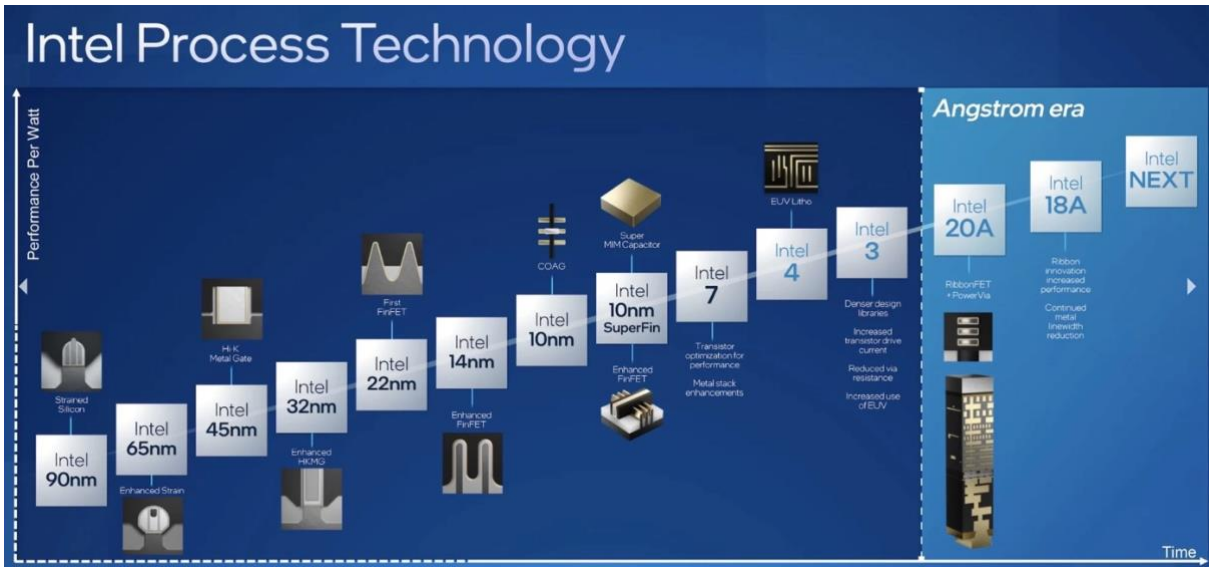
(TWI, 2021)

Appendix 3: The Chasm theory in the product life cycle



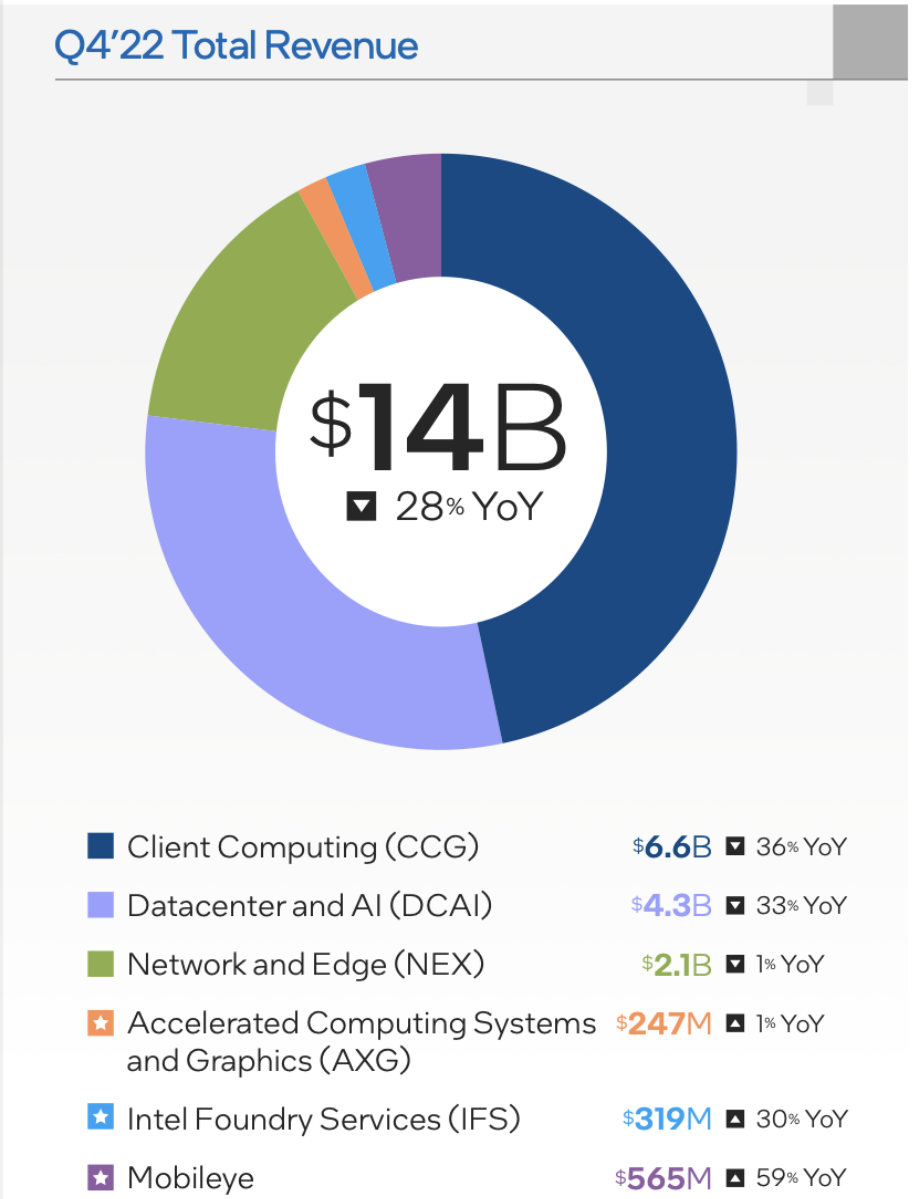
(De Bruin, 2020)

Appendix 4: Intel process roadmap



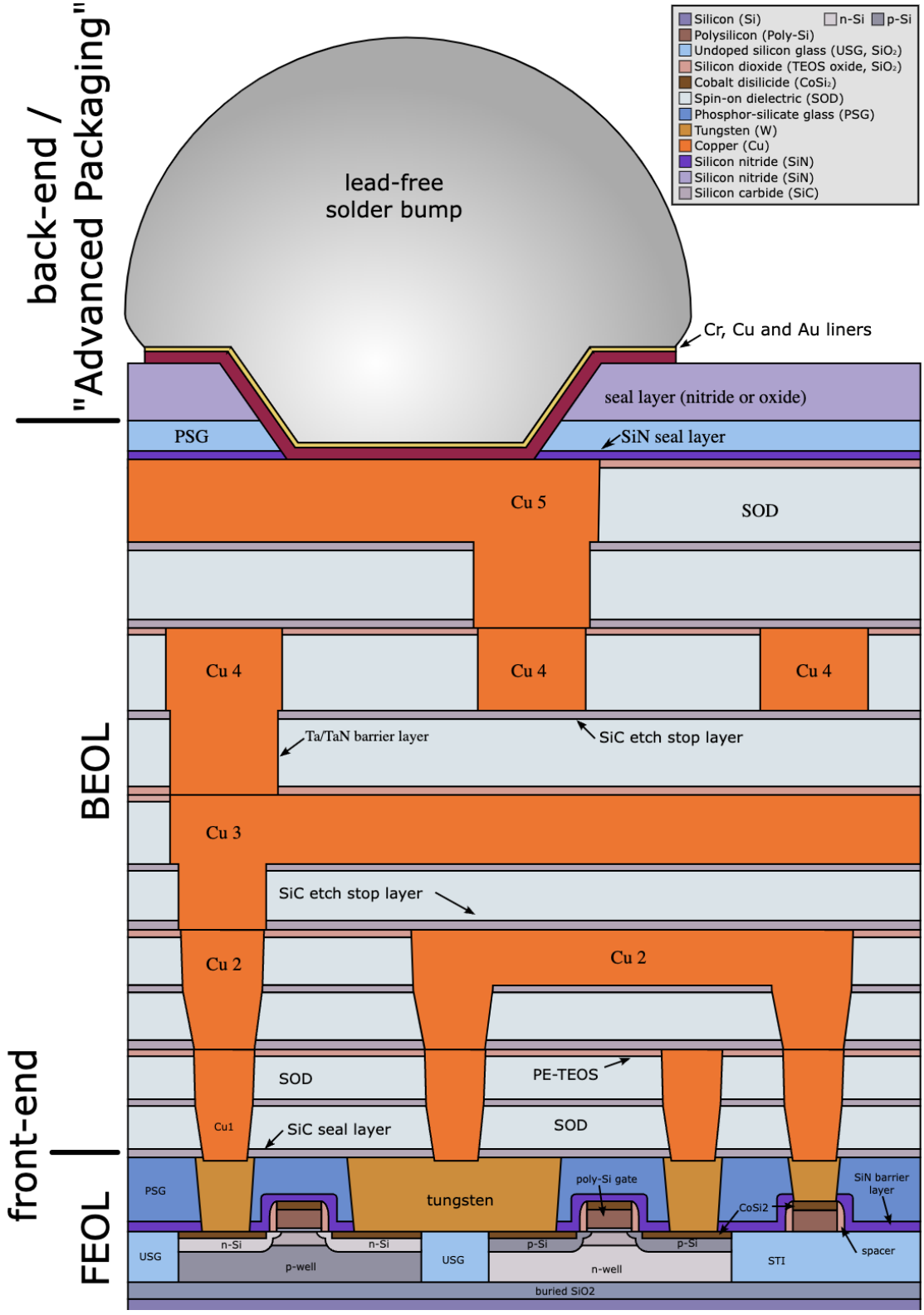
(Intel.com, 2024)

Appendix 5: Intel's Q4 2022 revenue per segment



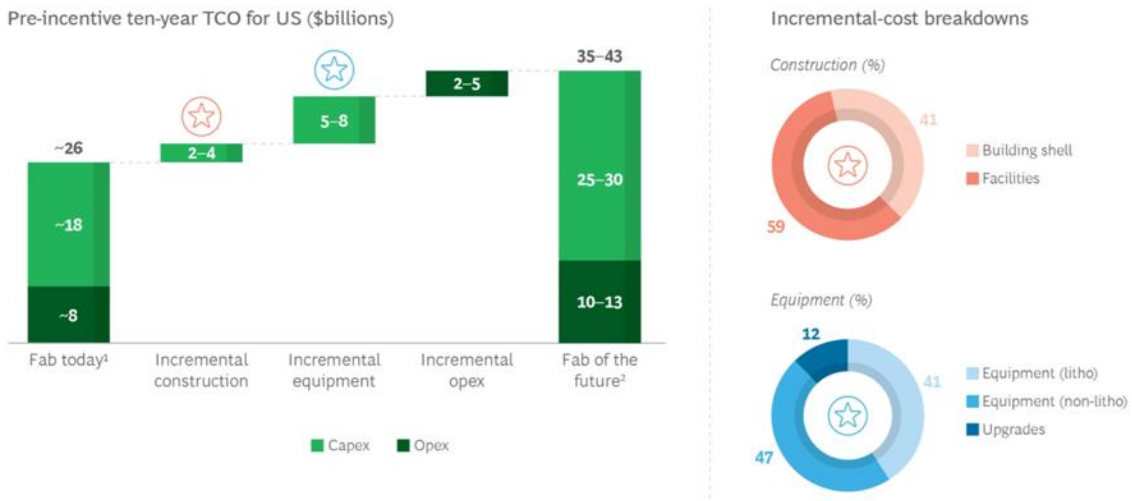
(" Intel Quarterly Financial and Business Report, n.d.)

Appendix 6: A chip's complex layers



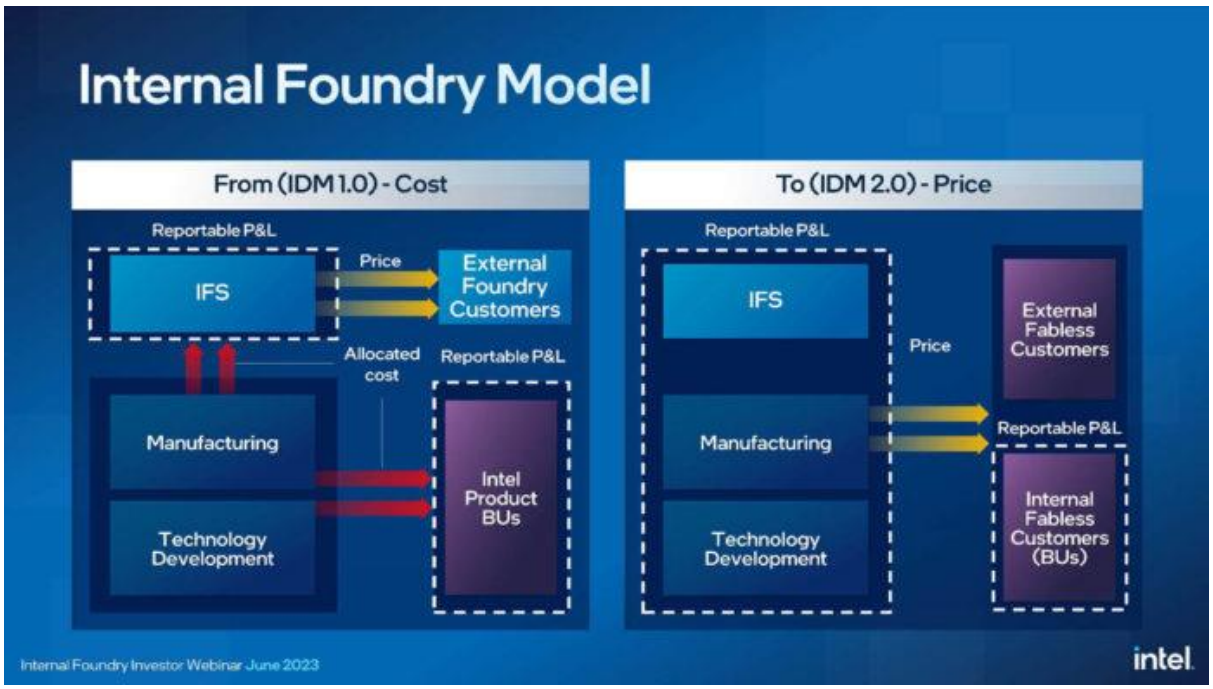
(Cepheiden, 2006)

Appendix 7: Advancements in fab technology and increased processing drive future fab costs even higher.



(BCG Global, 2023)

Appendix 8: Intel IDM1.0 versus IDM2.0 (Business Model)



(Intel IFS update, 2023)

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